

X2. Vary VR1 in the control section and check whether the voltage output at the secondary of X2 varies. A CRO can be used to observe this secondary voltage. It should be 50Hz sine wave, but if it has a break, it means the half cycles are not synchronised.

In Fig. 1, at point E, we have included a lag circuit comprising variable resistor VR2 (5-kilo-ohm) and capacitor C5 (10 μ F). Adjust preset VR2 until the waveform is a smooth sine wave.

There may be small ripples in the first half of each cycle, but these do not matter and will anyway be present due to PWM switching. Capacitor C7 (2.5 μ F) across the primary winding of transformer X2 filters them out.

The secondary voltage of transformer X2 should decrease and then increase as VR1 is raised from 0V position. Then check voltage regulation after changing over to auto position in Fig. 1. Adjust VR1 to the centre position precisely. In the centre position, there will be no pulse and therefore no adding voltage in the secondary winding. So the value of the zener diode used in the rectifier circuit should be changed in order to get 0V for 220V input. A variac is useful for varying the input voltage and checking the output.

5. Activate the buck-and-boost circuit by closing stabiliser switch S2 (Fig. 2).

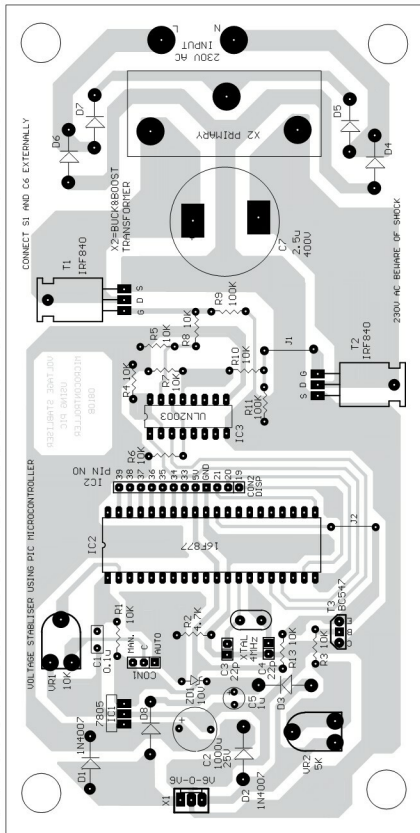


Fig. 7: Component layout for the PCB

Using a variac, the voltage can be varied and the stabiliser output observed on a voltmeter. If the voltage boosts up instead of bucking, reverse the secondary winding terminals connected in series to the mains.

Capacitor C6 (0.1 μ F) at the output terminals of X2 removes minor ripples, if any, in the waveform.

The optional input voltage display circuit consists of three common-anode, seven-segment LEDs (each LTS542) shown in Fig. 5. The seven-segment LED displays are driven from port-B of the chip in a multiplexed manner. The anode selections are made through bits 0 through 2 from port-D via transistors T4 through T6, respectively. Since the optional circuit was not tested at EFY Lab, its components are not included in parts list and the PCB layout. However, a 12-pin connector (CON2) has been provided on the PCB for extending the connections to the display circuit.

Construction

An actual-size, single-side PCB layout for the PIC microcontroller-based voltage stabiliser is shown in Fig. 6 and its component layout in Fig. 7.

Place the components on the PCB as per the circuits shown in Figs 1 and 2 and solder them. Give the mains AC power supply to the circuit through a variac and observe the stabilised output by varying the variac (auto transformer).

EFY note. The complete project folder containing the source codes

and other related files is included in this month's EFY-CD and is also available on www.efymag.com ●

The authors acknowledge the help of Mr Ramakrishnan, Sundaram Electronics, in designing this circuit